

Name: AYUSH RANA

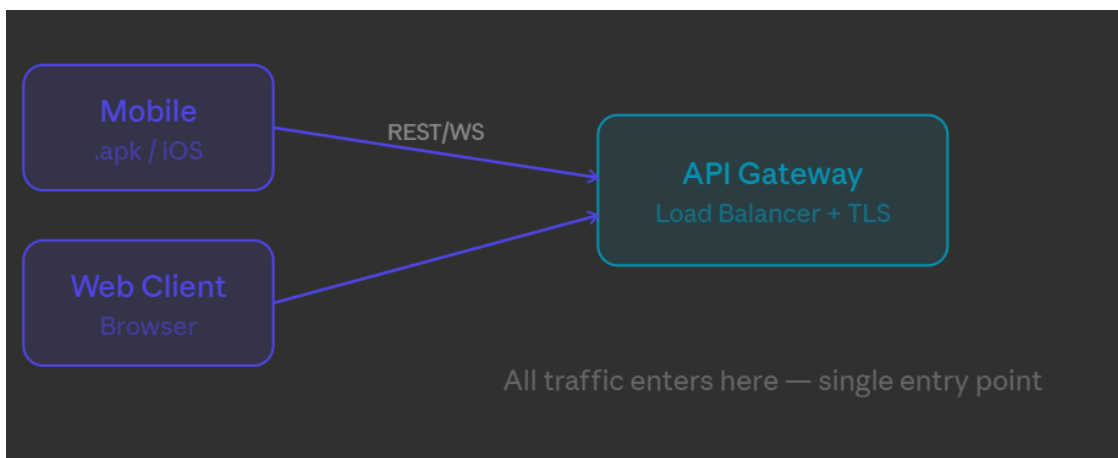
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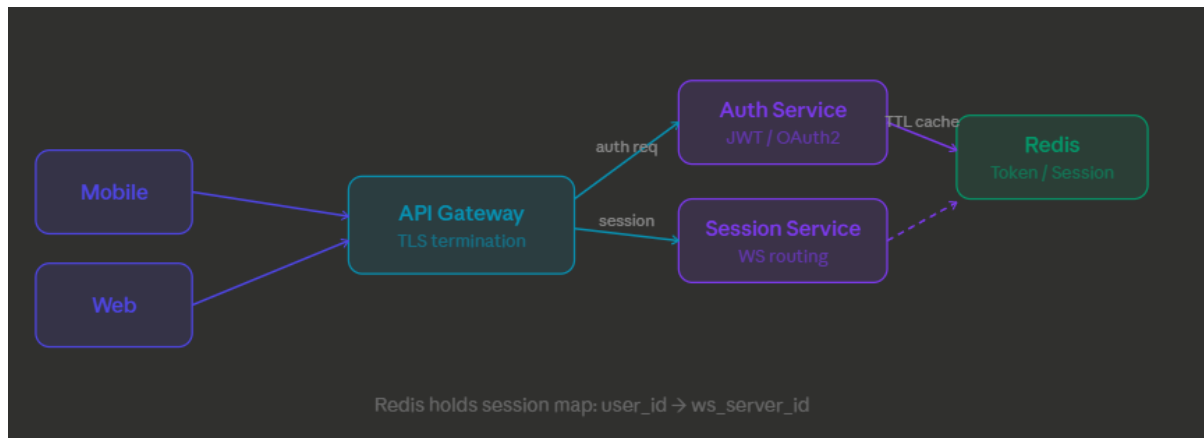
DESIGN CHAT APPLICATION

System design:-

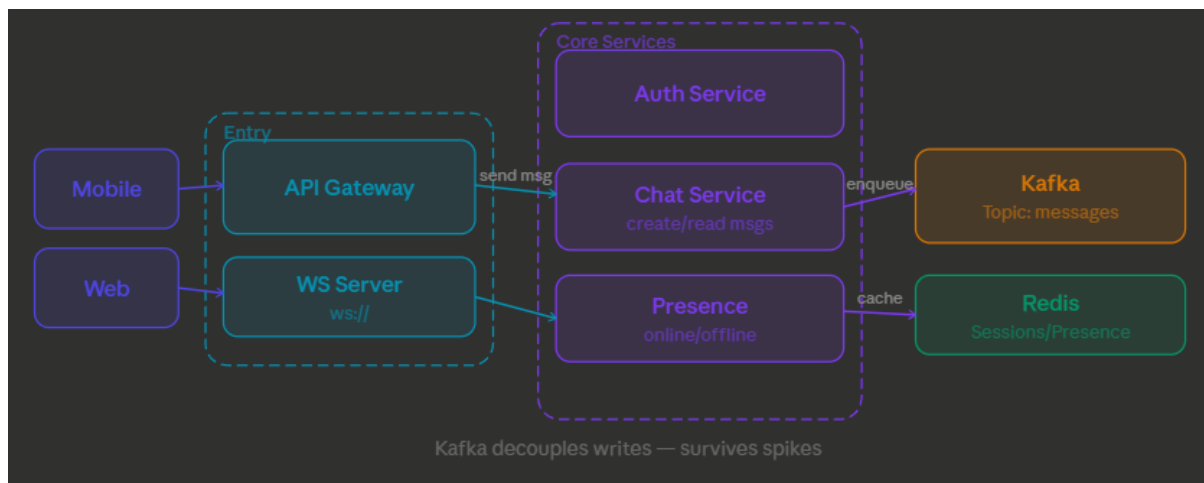
Start with the user-facing layer: mobile and web clients connect through a single API Gateway. This is the entry point for all communication REST for auth/profile, WebSocket for real-time messaging.



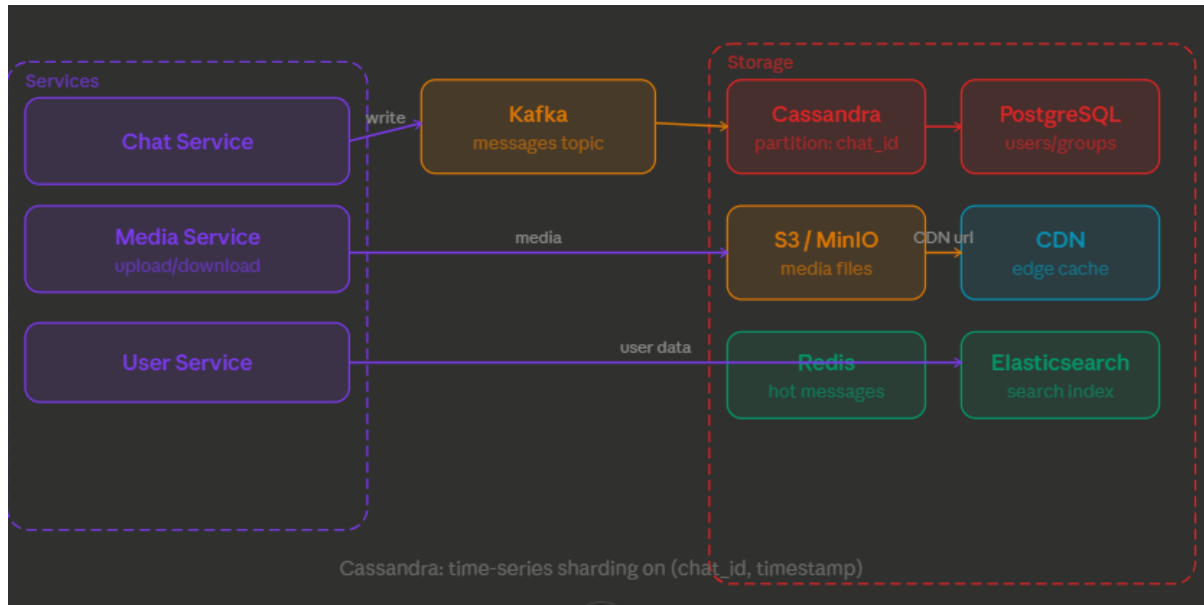
Add Auth Service for JWT token issuance and validation. Session Store (Redis) holds active WebSocket sessions mapping user → server node. Presence service tracks online/offline status.



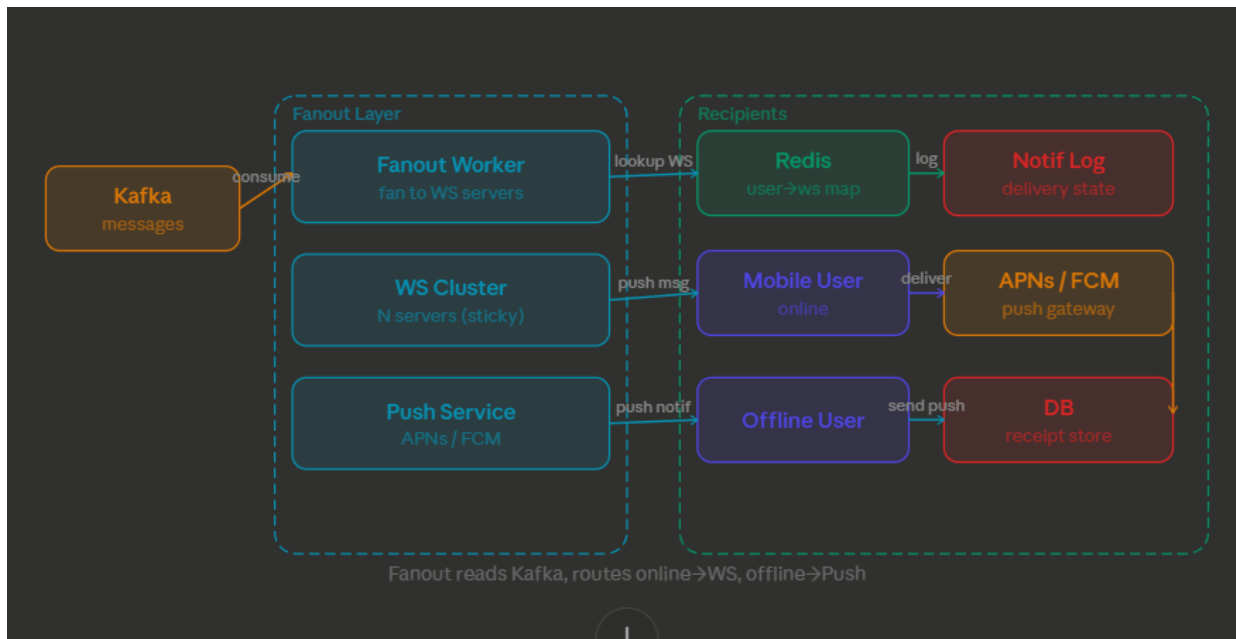
The Chat Service handles message creation and delivery. Message Queue (Kafka) decouples sends from storage writes ensuring no message is lost under load. WebSocket servers maintain persistent connections.



Messages stored in Cassandra (append-heavy, time-series partitioned by conversation_id). User/profile data in PostgreSQL. Media (images, files) in Object Storage (S3) with CDN for fast delivery



Push Notification Service fans out to APNs/FCM for offline users. Fanout worker reads from Kafka and delivers messages to all WebSocket servers that hold connections for recipient users.



Full architecture: load-balanced entry, stateless microservices, WebSocket cluster with session routing, Kafka fanout, Cassandra message store, Redis presence cache, S3 media, push notifications, and monitoring.

